

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.	(a)			Error 1 student is incorrect since solid weighed by difference (1) Error 2 Student is correct - improve by rinsing out the beaker (with distilled water) and transferring washings to the volumetric flask (before making up to 250 cm ³) (1)			2	2		2
	(b)			to show the end point / when to stop adding acid / when solution is neutralised neutral answer – to see a colour change	1			1		1
	(c)			award (1) for any of following so that a certain volume of acid can be added quickly before adding drop by drop to save time before doing accurate titrations to give a rough idea of the end point	1			1		1
	(d)			titres that are within 0.20 cm ³ of each other / within a small range / very close to each other	1			1		1
	(e)			Titration 2 final reading 23.65, initial reading 0.40 (1) Titration 3 final reading 23.90, initial reading 0.30 (1) mean titre = 23.60 cm ³ (1)		2	1	3		3
	(f)	(i)		$n(\text{HCl}) = 0.1 \times \frac{23.60}{1000} = 2.36 \times 10^{-3}$ (1) $n(\text{M}_2\text{CO}_3) = \frac{2.36 \times 10^{-3}}{2} = 1.18 \times 10^{-3}$ (1)		2		2	1	

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		(ii)		mass of carbonate = 1.25 (1) $n(\text{M}_2\text{CO}_3)$ in solution = $1.18 \times 10^{-3} \times 10 = 1.18 \times 10^{-2}$ (1) $M_r = \frac{1.25}{1.18 \times 10^{-2}} = 106$ (1) $A_r(\text{M}) = \frac{106-60}{2} = 23 \Rightarrow \text{M is sodium}$ (1) do not accept sodium if there is insufficient working to arrive at that conclusion ecf possible from part (i)		2	2	4	2	
				Question 12 total	3	6	5	14	3	8
				Paper total	28	35	17	80	20	13